

Energy-Aware Job Profiling and Mediation for Cloud and 6G Orchestration

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Abstract—Large-scale AI and cloud workloads are driving up data center energy demand, motivating orchestration that is explicitly energy-aware. This paper presents a practical methodology to characterize and predict job-level energy usage by linking publicly released Google traces: the 2019 Borg workload traces (per-instance CPU/memory usage and job metadata) and co-timed PowerData traces (five-minute power utilization at the power-distribution-unit level). We filter machines by power domain, aggregate per-instance usage into aligned five-minute windows, and construct job-level energy proxies that combine average CPU intensity, runtime, and coincident PDU utilization to derive low/medium/high energy classes. We then train lightweight classifiers (Random Forest, XGBoost) to predict a job’s energy class from pre-scheduling usage statistics and runtime, providing a mediation signal that an orchestrator can use to stagger high-energy jobs and backfill with low-energy ones to reduce peaks. The approach delivers two building blocks for sustainable orchestration aligned with 6G systems: empirical energy-job profiling and predictive mediation, grounded in real traces and implementable with modest complexity.

Index Terms—6G, energy-aware scheduling, workload profiling, data center power, Google cluster traces, orchestration, green computing

I. INTRODUCTION

Energy consumption in large-scale compute infrastructures has become a first-order design concern for cost, grid reliability, and sustainability. Recent analyses estimate data center electricity use in the hundreds of TWh annually [1], with growth increasingly driven by AI and cloud-native workloads [2]. Aligning compute demand with power availability—and ultimately with carbon-free energy—is therefore an urgent systems challenge [3], [4].

In parallel, the evolution toward Sixth-Generation (6G) systems envisages deep convergence of networking and cloud/edge computing. Energy efficiency and carbon awareness are repeatedly highlighted as key pillars for 6G, spanning *where* workloads execute (edge/core) and *when* they run relative to local power conditions [5]–[7]. Practical progress requires empirical methods that characterize job energy impact and provide signals to energy-aware orchestrators.

Public traces from production clusters provide a unique opportunity. Google’s Borg traces expose job/instance lifecycles, priorities, and per-instance usage; the 2019 May trace spans eight cells with normalized CPU and memory measurements

[8]–[10]. Complementing this, Google’s recently released *PowerData* traces provide five-minute power utilization at the Power Distribution Unit (PDU) level for the same month across multiple cells and domains, including data centers with a new Medium Voltage Power Plane (MVPP) [11], [12]. Together, these datasets enable—for the first time, to our knowledge—a study that links workload usage to measured facility power at meaningful spatial (PDU) and temporal (5 min) granularity.

Prior research has mined Google’s cluster traces to study scheduling, queuing, failures, and performance [9]. Separately, green/cloud literature proposes energy-aware scheduling policies and predictive models [13]–[15]. However, we find little work that *empirically* joins job/instance usage with *measured* power at PDU level to derive job energy classes and train predictive models that can inform scheduling. Most carbon-aware efforts use exogenous grid signals (e.g., carbon intensity) rather than facility power traces co-timed with the workloads [4], [16].

We empirically link Google’s Borg (workload) traces with co-timed PowerData (facility power) traces to construct and predict job-level energy profiles. Specifically, we:

- Develop a preprocessing pipeline that flattens per-instance usage, merges with job metadata, and filters by machine→PDU mapping to align workload activity with PDU power utilization.
- Construct job-level energy proxies that capture both intensity (CPU usage) and duration, and label jobs into *low/medium/high* energy classes using distributional thresholds.
- Train and evaluate classifiers (Logistic Regression, Random Forest, XGBoost) to predict job energy class from pre-scheduling features (usage statistics, runtime, priority, scheduling class).
- Conduct a drop-one-feature ablation study to identify critical predictors, showing that `avg_cpus` and `runtime_sec` dominate, while memory and efficiency metrics have marginal effect.

Our key contributions are:

- 1) **Trace linkage methodology.** A reproducible pipeline that joins Borg usage traces with PDU-level PowerData,

enabling analysis of workload→power coupling at facility granularity.

- 2) **Energy-aware job profiling.** An empirical characterization of jobs into energy classes (LOW/MED/HIGH) using aligned workload and power windows.
- 3) **Predictive mediation.** Demonstration that lightweight classifiers, especially Random Forest and XGBoost, achieve near-perfect prediction of job energy class, providing a practical signal for orchestrators to stagger high-energy jobs and backfill with low-energy ones.
- 4) **Feature sensitivity analysis.** A systematic ablation study showing that CPU usage and runtime are decisive predictors, while other workload metrics add limited marginal value.
- 5) **6G relevance.** We discuss how these profiling and mediation signals can be integrated into energy-aware orchestration for 6G systems, where scheduling decisions must balance latency, cost, and sustainability.

By grounding energy-aware orchestration in empirical workload–power traces, our study provides a reusable foundation for sustainable scheduling in cloud/edge infrastructures envisioned for 6G systems. Subsequent sections present the methodology (§II), results (§III), and conclusions (§IV).

II. METHODOLOGY

We implement a two-layer pipeline that enables energy-aware orchestration: (i) *Energy–Job Profiling*, which empirically links workload usage to facility power at the Power Distribution Unit (PDU) level, and (ii) *Mediation Strategy*, which predicts a job’s energy class from workload features to inform scheduling. Our analysis combines Google’s 2019 Borg traces (v3) [8]–[10] with the co-timed PowerData traces [11].

A. Data Sources

Borg traces (v3). We use three tables: `instance_usage` (per-instance CPU/memory usage, 5 min windows, normalized units), `instance_events` (task→machine lifecycle mapping), and `collection_events` (job metadata including priority and scheduling class). **PowerData.** We use per-PDU, 5 min *measured_power_util* and the machine→PDU mapping table, which specifies which PDU and cell power a given machine.

B. Data Preparation

We implemented a preprocessing pipeline as follows:

- *Flattening.* Each `instance_usage-*.json.gz` shard is decompressed and flattened to extract numeric fields: average CPU/memory usage, maximum usage, assigned and page-cache memory, cycles per instruction, and memory accesses per instruction.
- *Merging.* The flattened usage is joined with `instance_events` to resolve instance→job mappings, and with `collection_events` to obtain job metadata (priority, scheduling class).

- *Filtering.* Using the machine→PDU mapping, we restrict analysis to *Cell A*, *PDU6*. Only instances running on machines connected to this PDU are retained.

C. Job-Level Energy Summaries

For each job j with instances $\mathcal{I}_j$, we compute:

$$C_j = \sum_{i \in \mathcal{I}_j} \sum_{b \in \mathcal{B}_j} u_{i,b} \cdot \Delta, \quad (1)$$

$$\bar{u}_j = \frac{1}{|\mathcal{B}_j|} \sum_{b \in \mathcal{B}_j} \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} u_{i,b}, \quad (2)$$

$$R_j = \max_t(\text{end}_t) - \min_t(\text{start}_t), \quad (3)$$

where $u_{i,b}$ is average CPU usage of instance i in bin b , $\Delta = 300$ s, and $\mathcal{B}_j$ are bins where j is active. The *energy proxy* is $E_j = C_j \times R_j$, capturing both intensity and duration, consistent with proxy-based approaches to energy modeling in cloud data centers [15].

D. Energy Class Labeling

Jobs are partitioned into *low*, *medium*, and *high* classes using tertiles of E_j , a simple distributional thresholding also used in prior energy-aware scheduling studies [17]:

$$\text{label}(j) = \begin{cases} \text{LOW}, & E_j \leq Q_{1/3}, \\ \text{MED}, & Q_{1/3} < E_j \leq Q_{2/3}, \\ \text{HIGH}, & E_j > Q_{2/3}, \end{cases} \quad (4)$$

where $Q_{1/3}$ and $Q_{2/3}$ are the 33rd and 66th percentiles across all jobs.

E. Feature Set

Each job j is represented by:

- *Usage features:* average CPU ($\bar{u}_j$), average memory, assigned memory, page cache memory.
- *Efficiency metrics:* cycles per instruction, memory accesses per instruction.
- *Temporal:* runtime R_j .

F. Predictive Models

We evaluate multinomial Logistic Regression, Random Forest, and XGBoost. Labels are encoded as $\{0, 1, 2\}$ for $\{\text{LOW}, \text{MED}, \text{HIGH}\}$. Training uses 5-fold cross-validation, stratified by energy class. Metrics reported are accuracy, macro-F1, confusion matrix, and feature importance (tree-based).

G. Ablation Study

To assess feature relevance, we perform a drop-one-feature ablation: models are retrained with one feature removed, and cross-validation accuracy is compared with the baseline (all features). This identifies the most critical features for predictive performance.

III. RESULTS

We summarize workload–power correlations, energy class distribution, classifier performance, and feature ablation.

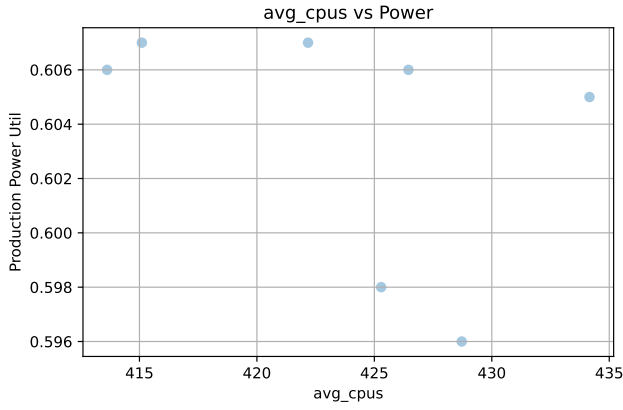


Fig. 1. Aggregated CPU usage vs. PDU measured power utilization for Cell A, PDU6.

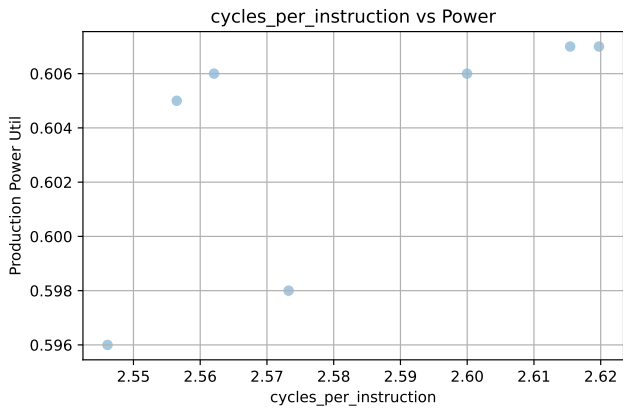


Fig. 2. Cycles per instruction vs. PDU measured power utilization for Cell A, PDU6.

A. Workload–Power Alignment

We analyze correlations between multiple workload features and measured PDU power utilization. Aggregated instance CPU usage per 5 min bin (U_b) shows a clear positive correlation with PDU measured power (P_b), confirming CPU load as the primary driver of IT energy demand. In contrast, efficiency metrics such as cycles per instruction and memory-related features such as page cache memory exhibit little to no correlation with PDU power, reflecting the influence of cooling and non-IT loads at facility level. Figs. 1–3 illustrate these relationships.

B. Energy Class Distribution

Fig. 4 shows the distribution of job-level energy estimates E_j . The distribution is skewed, with many low-energy jobs and fewer high-energy outliers. To ensure balanced learning, we apply tertile-based thresholds (33rd, 66th percentiles), resulting in approximately equal proportions of Low, Medium, and High jobs. This labeling strategy captures the heavy-tail nature of workloads while maintaining class balance for classification [17].

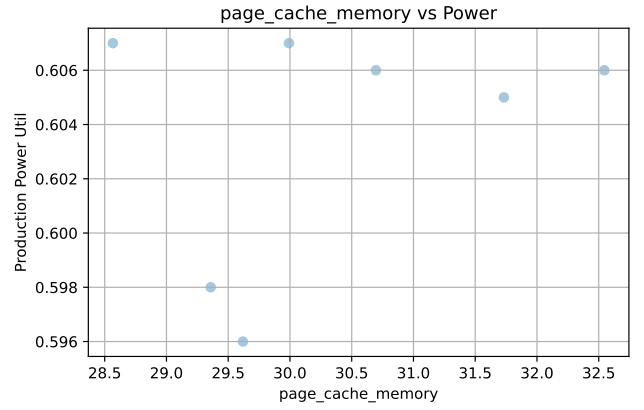


Fig. 3. Page cache memory vs. PDU measured power utilization for Cell A, PDU6.

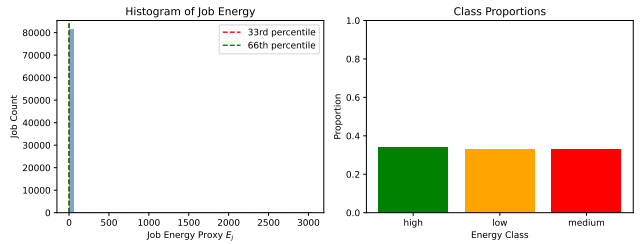


Fig. 4. Distribution of job-level energy proxy E_j and resulting Low/Medium/High classes.

C. Classifier Performance

Table I reports test-set metrics. Tree ensembles significantly outperform logistic regression.

Confusion matrices (Figures 5–7) show that while Logistic Regression misclassifies across all classes, Random Forest and XGBoost achieve nearly perfect separation.

D. Feature Importance

Fig. 8 shows feature importance for Random Forest and XGBoost. Both models identify `avg_cpus` and `runtime_sec` as dominant predictors, while memory and efficiency metrics contribute marginally.

E. Ablation Study

Drop-one-feature analysis (Fig. 9) confirms `avg_cpus` and `runtime_sec` are essential: removing them reduces accuracy from 0.996 to 0.84–0.90. Other features have minimal impact.

F. From Mediation to Orchestration (Illustration)

To illustrate practical impact, we sketch how predicted energy classes (LOW/MED/HIGH) can guide orchestration. A scheduler can defer a subset of HIGH-energy jobs and backfill with LOW-energy ones, thereby flattening peaks in aggregate load while keeping throughput unchanged [4], [16]. This demonstrates how *mediation signals* from classification can be integrated into orchestration decisions. Fig. 10 provides a schematic view.

TABLE I
ENERGY CLASS PREDICTION ON HELD-OUT TEST SET.

Model	Accuracy	Macro-F1
Logistic Regression	0.784	0.787
Random Forest	0.994	0.994
XGBoost	0.996	0.996

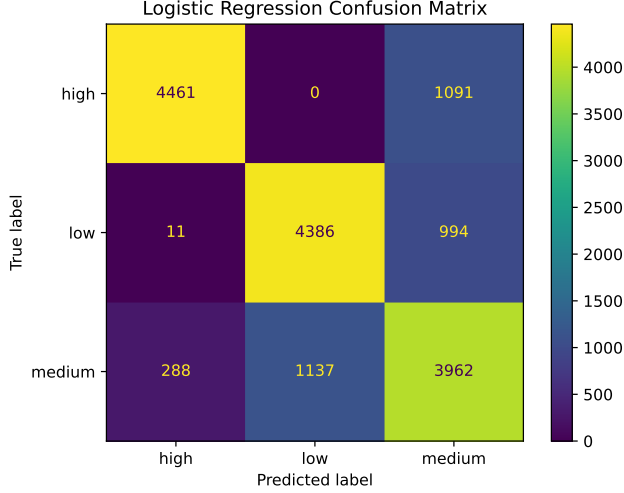


Fig. 5. Confusion matrix for Logistic Regression.

IV. CONCLUSION AND FUTURE WORK

We presented a practical methodology that links Google’s Borg (v3) workload traces with co-timed PDU-level Power-Data to enable *Energy–Job Profiling* and a *Mediation Strategy* for energy-aware orchestration. Concretely, we (i) filtered machines by power domain, (ii) aggregated instance CPU usage into aligned 5 min windows, (iii) built job-level energy summaries and LOW/MED/HIGH labels, and (iv) trained lightweight classifiers to predict a job’s energy class from pre-scheduling features. Empirically, we observed a consistent workload–power alignment, particularly between aggregate CPU and measured PDU utilization, and our predictors provide an actionable mediation signal that can be used to stagger high-energy jobs while backfilling with low-energy ones, thereby reducing peak demand without changing total work.

From a systems perspective, the profiling and mediation signals form reusable building blocks for energy-aware orchestration in cloud/edge infrastructures anticipated in 6G systems, where scheduling decisions must consider not only latency and cost but also local power conditions and sustainability goals. Because our approach relies on publicly available traces and a transparent data-processing pipeline, it can serve as a reference for subsequent studies that require empirical grounding.

Our analysis operates at PDU granularity and inherits trace characteristics (normalized resource units, missing/synthesized events, and inclusion of non-IT loads in power utilization). We evaluated on a single cell/domain initially, and labels were defined via distributional tertiles rather than absolute

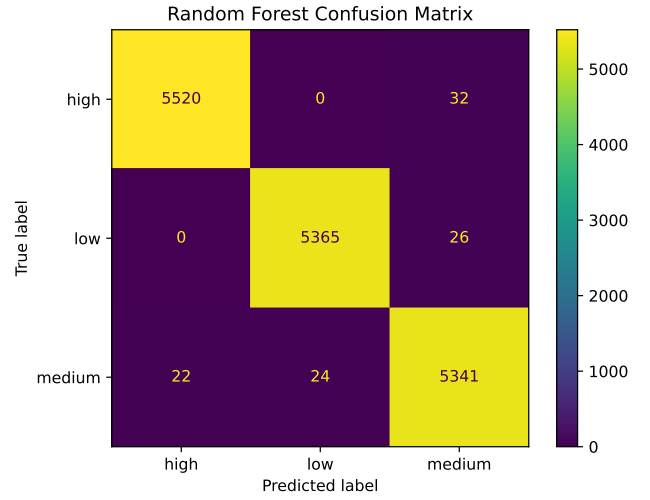


Fig. 6. Confusion matrix for Random Forest.

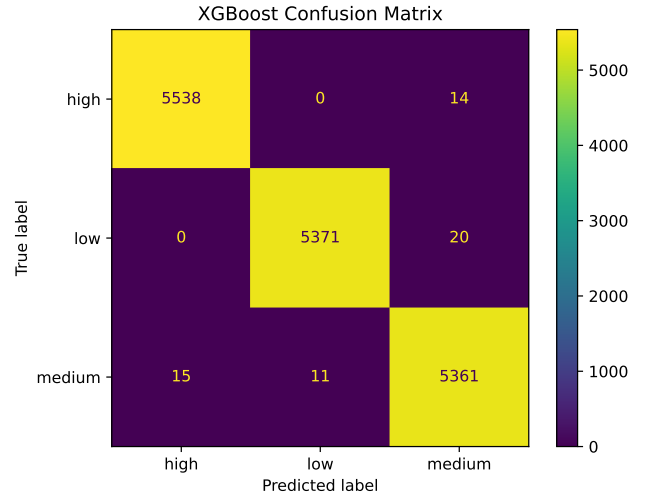


Fig. 7. Confusion matrix for XGBoost.

energy thresholds. These choices emphasize feasibility and reproducibility for a short letter.

We will (i) generalize across multiple PDUs and cells and report cross-site robustness; (ii) refine labels by combining co-incident PDU power with production-only utilization and absolute energy normalization; (iii) incorporate exogenous signals (e.g., renewable availability, carbon intensity, and price) and uncertainty estimates to improve mediation quality; (iv) extend mediation to full orchestration (run/delay/fallback, admission control, scaling) and evaluate peak-shaving vs. deadline adherence; (v) explore learning-to-schedule (contextual bandits/RL) with safety constraints and fairness to priority classes [18]; and (vi) assess applicability in telecom-edge scenarios aligned with 6G, including co-optimization of compute and RAN power budgets [19], [20]. We plan to release code snippets for the trace-linkage pipeline to facilitate reproducibility and community uptake.

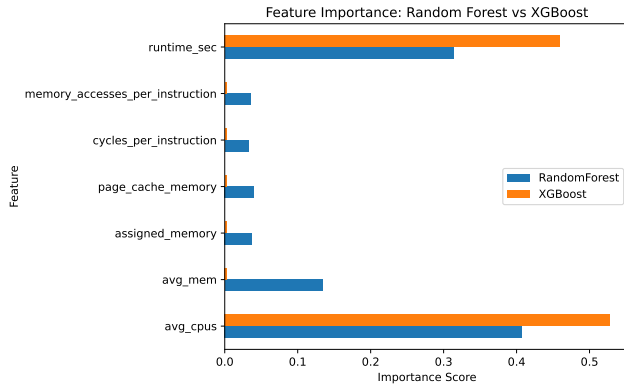


Fig. 8. Feature importance (Random Forest vs XGBoost).

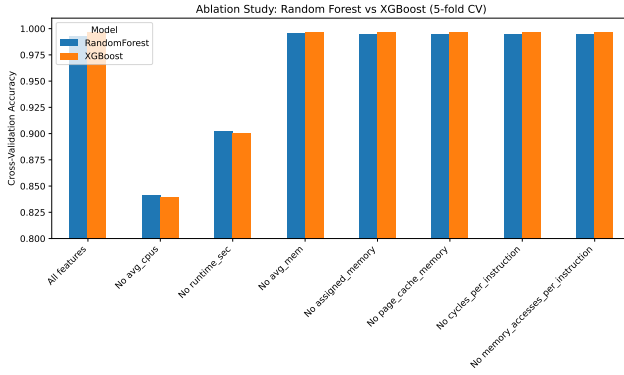


Fig. 9. Ablation study comparing Random Forest and XGBoost.

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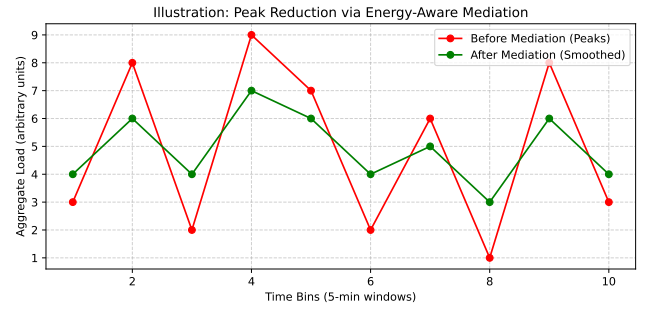


Fig. 10. Conceptual illustration of energy-aware mediation. By deferring some HIGH-energy jobs and backfilling with LOW-energy jobs, peak aggregate load is reduced while total work remains constant.

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